

ARM CORTEX-M3 POWERED SMART HELMET FOR ACCIDENT DETECTION

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Abstract—Road accidents and deaths due to these have become exceptionally common across the world, claiming millions of lives. Road accidents have long been viewed as unavoidable, but the reality is that they are preventable tragedies affecting countless lives. According to the World Health Organization (WHO), road crashes remain one of the leading causes of death globally, leaving families shattered and communities grieving. Every accident represents a person—a life full of dreams, relationships, and aspirations—that was suddenly disrupted. Previously, it was said that road accidents are inevitable, unpredictable, and non-preventable, especially in the case of two-wheeler accidents. However, road accidents and their deaths are avoidable, specifically. Too many riders lose their lives because of missing helmets or delayed medical help. Simple precautions can save lives and bring loved ones home safely. Even after implementing strict traffic rules, the riders avoid wearing helmets due to negligence or discomfort. New technology has to come into the frame, so our thesis progresses in that direction. Integrating embedded systems into the helmet to innovate a smart helmet that provides real-time alerts to riders, preventing harm while encouraging helmet use. This innovative solution seeks to improve the safety of motorcyclists by integrating accident detection and alert systems. The Smart Helmet utilizes advanced sensors, microcontrollers, and various communication modules to monitor the rider's environment and detect potential threats. Upon detecting any accident, the system actively sends alerts to the rider, ensuring a swift response that can be fundamental in saving lives. The proposed work points to the details about the conceptual framework, methodology, and implementation process of the system. It also presents simulation results demonstrating the effectiveness of the helmet in real-world scenarios. Additionally, the future scope of the proposed work might highlight essential enhancements and broader applications. The Smart Helmet offers significant societal benefits, representing a promising advancement in motorcycle safety technology.

Keywords—Road accidents; Embedded System; accident detection; Smart Helmet; Motorcyclists safety; Microcontrollers; sensors; real-time alerts.

I. INTRODUCTION

The quick changes in technology have assisted us in revolutionizing. Every perspective is shaped by experiences, helping us connect and understand each other. We are now focusing on safety and convenience for our mobility and leisure activities daily. Of all the trendy gadgets that have come to prominence in recent years, the smart helmet is an excellent example of the combination of advanced technology and essential safety measures. This technical report delves into the complex universe of smart helmets and explores their design, functionality, and possible uses in many areas.

The Smart helmets are designed to provide improved safety and experience to users with advanced characteristics like sensors, connectivity options, and real-time processing. Smart helmets are used as safety equipment, along with advanced peripherals meant to keep track and ensure safety and enhance the health and well-being of the wearer. Bluetooth connectivity, GPS, and impact sensors convert the traditional helmet into a smart, connected device with unmatched advantages.

The strongest aspect of smart helmet functionality is the utilization of embedded systems. Embedded systems refer to systems that are embedded in larger systems for the execution of specific sets of functions, and they are at the heart of smart helmet functionality. Embedded systems are composed of microcontrollers, sensors, and actuators that communicate internally to handle data and carry out specific processes. Embedded systems, in the case of smart helmets, provide for real-time observation of the surroundings and physical state of the wearer and offer them the key information that can be used to improve safety and performance.

The focus of this paper is to provide a detailed description of the smart helmet, its different components, technological advancements, and why this one is better than the conventional ones. In addition, this seeks to identify various applications of smart helmets across various sectors, including sports and motorcycling, cycling, construction, and sports and hence emphasize their versatility and far-reaching nature. The proposed work will also touch on the application of embedded systems in these helmets and how they help enhance the use of the helmets.

With the world growing more connected each day, the need for smart, wearable technology increases day by day. Smart helmets are one such phenomenon, explaining how technology is woven into everyday objects in a bid to maximize safety, ease, and overall user experience. Embedded systems are at the frontline of this integration, making smart helmets provide such services as collision detection, biometric sensing, and voice-controlled directions. Such systems allow the helmet to gather and process information from other sensors and send feedback and notifications to the users.

The application of ARM Cortex-M3 microcontrollers to smart helmets is a turning point for accident detection technology. ARM Cortex-M3 microcontrollers are distinguished by their high performance and efficiency, which qualify them for real-time data processing and analysis. With the implementation of advanced sensor fusion algorithms, the smart helmet can efficiently detect and respond to impact events, giving timely notification and intervention. The ARM Cortex-M3's Transistors facilitate seamless integration with numerous sensors, such as accelerometers and gyroscopes, to effectively monitor the helmet's orientation and motion. This advanced system provides higher reliability and accuracy in accident detection, providing an essential safeguard for motorcyclists. Additionally, the ARM Cortex-M3's low power consumption offers longer battery life, making the smart helmet convenient and reliable for everyday use.

This paper will examine the technical capabilities, interface, and real-world applications of smart helmets, offering a glance at what they are capable of now and what they will be able to do in the future. Through researching the role of embedded systems, we will gain a better idea of how these technological wonders work and what advantages they provide for their users.

At the conclusion, readers will gain a comprehensive grasp of the functionality, advantages, and innovative technology behind the smart helmet and how it has become an indispensable part of current safety equipment. The Smart Helmet is a monumental step in individual safety and connectivity, and has the potential to revolutionize our perception of head protection and wearable technology.

II. LITERATURE SURVEY

In this research work [1], the author proposes a smart helmet to safeguard the situation of road accidents and the fatalities occurring due to the absence of mind and negligent attitude displayed by the riders while riding, prevalent in the Asian continent. Due to this reason, the mode of work reflects a systematic helmet embedded with preliminary conditions that are required for the motorbike to move. The most striking concepts proposed in the work are compulsory use of helmets, avoiding all forms of drug and addiction activities, and the use of side stands before the engine of the vehicle is ignited; avoiding any of these conditions would not initiate the engine. When an accident or any important situation is sensed, the system alerts through SMS to the predefined emergency contact. The prototype runs on different modes such as initial mode, moving mode and emergency mode, each of which comprises different functionalities. The rider must navigate through every mode for the whole vehicle system to function effectively. The process of operation of such modes as described in the work paper is as follows: initially, the rider needs to overcome the initial mode to the light system, continuous monitoring of the bike and rider while the moving mode is started. In any unexpected conditions during moving mode, the advanced system keeps its foot in emergency mode, enabling the alert module system to send an alert message to the emergency number.

The concept put forth in this paper is to integrate the hardware system with the cloud computing system for the detection and reporting purposes in case of an emergency [2]. The casualty detection system keeps on tracking the surroundings of the rider and also offers accelerometer values to the processing system for abnormal fluctuations. In the event of an accident, the pertinent information is sent to the emergency contacts through a cloud-based service.

Vehicle location is accessed using the global positioning system. The system provides real-time and reliable sending of information concerning the accident, which gets updated into the computing software infrastructure that is accessed by IOT. Accordingly, based on the pervasive connection that is a strong attribute for the high-tech cities, a smart helmet accident detection system is introduced. The use of INFORMATION & COMMUNICATION TECHNOLOGIES (ICTs) has significantly advanced Occupational Health and Safety (OHS) by enhancing workplace safety.

Personal Protective Equipment (PPE) constructed with ICTs mitigates accidents in the workplace because it can make decisions based on sensing environmental states [3]. Paradigms like the Industrial Internet of Things (IIoT) and Artificial Intelligence (AI) enable the development of advanced PPE with features like environmental sensing, threat detection, and warning mechanisms. These systems constantly monitor workplace environments, notifying workers and managers of threats. This paper introduces a prototype smart helmet that can evaluate near-real-time environmental risk. The sensor data is sent to an AI platform for processing. There were 11,755 samples in 12 scenarios in the training dataset. A comparison of supervised models was conducted, and a Deep Convolutional Neural Network (CNN) for occupational danger detection was recommended. After preparing data for CNN compatibility, the result showed cross-validation accuracy of 92.05%, superior to other traditional methods approaches such as Static Neural Networks, Naive Bayes Classifiers, and Support Vector Machines (SVM).

In this paper, the Author directs focus towards intoxicative driving and the deaths caused by that type of driving. According to Section 129 of the Motor Vehicles Act, 1988, all the passengers of a two-wheeler must wear head protective devices according to BIS (Bureau of Indian Standards) standards [4]. Additionally, drunken driving under the influence (DUI) is an offence according to the Motor Vehicle Act 1939, for which a punishment will be given to the driver. Bike riders can simply escape the law today. An additional concept is about an alcohol sensor being a breath analyser that identifies the amount of alcohol in the rider's breath, and if it is beyond the acceptable level, it cannot be ignited. It will send a message to the number that says "Rider is drunk and is attempting to drive the bike". The MQ-3 sensor is employed for it. The methodology proposed in this research work is that a Force Sensing Resistor[5] is inserted inside the helmet, where human touch is monitored.

This is ascertained by the helmet unit as to whether the helmet unit is on or off the rider's head prior to the bike's ignition. If the requirement is met, then it instructs the bike unit to start up. Force Sensing Resistors or FSRs are long-life polymer thick film (PTF) sensors whose force-sensitive resistance is lower with force applied to the sensor face. The sensor is utilized in human touch control in numerous applications including medical equipment, automotive electronics and industry and robotics. Force sensing resistor is an applied force variable resistance two-wire sensor.

III. METHODOLOGY

Imagine a world in which motorcycle accidents are fewer and farther between. One where motorcyclists can command the open road with confidence, knowing they're protected by the best technology.

That is a reality today, courtesy of our innovative Smart Helmet. This headband innovation uses advanced sensors and AI to detect accidents and alert emergency services in real time.

This system is intended to detect accidents and send alarms using an ARM Cortex-M3 microcontroller. There are a number of sensors in the helmet to monitor the rider's condition as well as external impact. If it experiences any lethal accident, this system gives an alert using a communication module to inform emergency contacts or services.

The smart helmet houses a variety of sensors that monitor the motion, body rhythms, and surroundings of the rider. Sensors use smooth algorithms that are capable of detecting even minute deviations.

If something happens, the helmet acts. It sends out an emergency call to emergency services with GPS coordinates and vital signs data. In this manner, assistance arrives as quickly as possible and is prepared to handle whatever they are up against.

Our Smart Helmet is powered by an ARM Cortex-M3 microcontroller, which offers the processing capability necessary for real-time analysis of sensor data. The helmet further comprises an array of sensors, namely:

- Gyroscope and accelerometer to sense movement and orientation
- GPS module to supply location information
- A buzzer goes off to alert nearby people.

SYSTEM DESIGN AND ARCHITECTURE

ARM Cortex-M3 Microcontroller (e.g., LPC1768) – Is the core unit for collecting and processing of data.

Accelerometer & Gyroscope (e.g., MPU6050) – Detect Sudden impacts, tilts, or falls.

GPS Module (e.g., Neo-6M) – Tracks the rider's live location & sends coordinates at times of any accident.

Battery Unit – Provides power to the entire system.

Buzzer – Alerts nearby people during an accident.



Fig 1. Block Diagram of the proposed work

SYSTEM IMPLEMENTATION

The accelerometer and gyroscope continually monitor the movement of the helmet.

The microcontroller receives continuous input from sensors. If a collision exceeds a threshold value (derived from experiment data), the system identifies it as a potential accident.

The system cross-verifies the accident by discovering the rider's vital signs (e.g., abnormal)

Once an accident has been confirmed, the GPS module obtains the coordinates of the site of the accident.

The GSM module sends an emergency message with the location of the accident to contacts.

A buzzer and an LED indicator are activated to notify people around.

SOFTWARE DEVELOPMENT

Programming Language: Embedded C/C++

Development Environment: Keil μ Vision for ARM Cortex-M3

Algorithm Development: Sensor calibration and data processing Threshold-based accident detection logic

TESTING AND VALIDATION

Unit Testing: We extensively test sensors so that they provide precise and reliable measurements.

Integration Testing: We test the whole system so that all the pieces of the system talk to each other seamlessly and work as expected.

Simulation Testing: Different accident scenarios are simulated for evaluating detection accuracy.

Field Testing: The helmet is field tested in order to test the reliability

This ARM Cortex-M3 smart helmet provides a smart method of quick accident identification and prompt response through the integration of impact, health, and communication modules. It facilitates quick emergency response, improving the chances of saving lives during emergencies.



Fig. 2 Flow diagram

IV. EXPERIMENTAL RESULTS AND ANALYSIS OF PROPOSED WORK

Our suggested ARM Cortex-M3-based smart helmet system experimentation results and analysis consider exploring its effectiveness in accident sensing through the MPU6050 and ST123004PA sensors. We tested the system to provide assurance that it is able to detect and tell apart what it is supposed to sense: rotational motion, abrupt impacts, and sensor response. The system's accuracy, response time, and reliability were compared to other available accident detection approaches. Critical parameters like acceleration changes, gyroscope data, and sensor data were compared to determine the effectiveness of the helmet. The findings are in agreement that our smart helmet offers a quicker and more precise detection system, enhancing riders' safety.

More comparisons put our system against past work, emphasising detection accuracy improvement and response time. MPU6050 and ST123004PA sensors accurately sense motion dynamics, limiting false alarms and enhancing detection accuracy. Comparative experiments with previous accident detection approaches demonstrate our system's enhanced performance, highlighting real-time alerting capabilities. The response time analysis verifies the helmet's effectiveness in dispatching emergency notifications in real time. In general, experimental trials validate our smart helmet as an effective safety device, integrating superior sensor technologies for real-world accident detection and prevention in motorcyclist safety applications

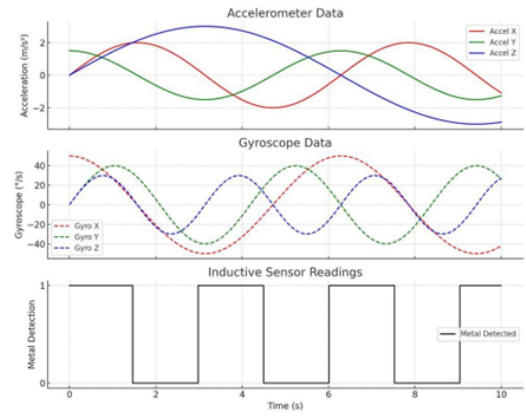


Fig. 3 Graphical analysis of the written code

The graph above displays accelerometer, gyroscope, and inductive sensor readings, which are critical for accident detection in the smart helmet system. The accelerometer data graph displays acceleration in the X, Y, and Z directions, which is useful for detecting impacts that happen randomly and indicate an accident. The gyroscope data graph displays rotational movement, which is useful to detect head tilts or sudden falls. The inductive sensor reading graph indicates metal detection, confirming helmet wear before launching the accident detection. The combination of these sensors improves accuracy through minimising false positives, real-time detection, and maximising safety for the rider. This multi-sensor setup guarantees effective accident detection and response.

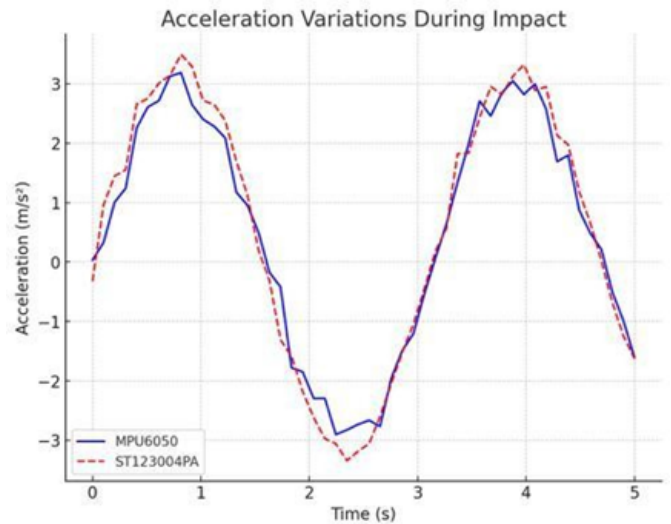


Fig 4. Variations in acceleration throughout the impact phase

The MPU6050 and ST123004PA sensors' accelerations are read in real time when an impact occurs. The y-axis is acceleration in meters per second squared (m/s^2), and the x-axis is seconds. The MPU6050 sensor (blue line) and the ST123004PA sensor (red dashed line) differ in their response profiles and therefore their sensitivity to respond to sharp motion changes. The collision points at the acceleration peak are the points that are most important in accident detection. The higher the value of acceleration, the higher the impact. The comparison between the sensors will determine which one responds faster and more accurately in accident detection. This calculation is important in building a smart helmet that can instantaneously detect and react to impacts.

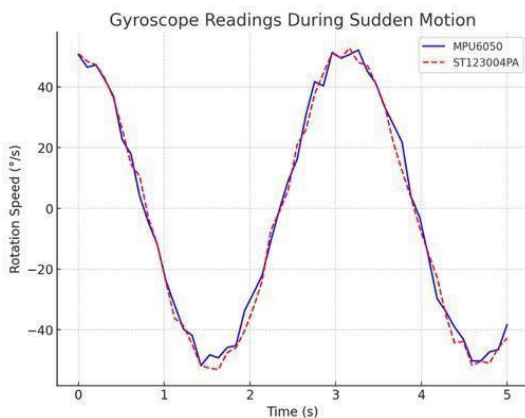


Fig. 5 Readings from the gyroscope

Comparison of MPU6050 and ST123004PA gyroscope outputs, capturing rotational motion in sequence, is shown. The x-axis is time, and the y-axis is rotational speed in degrees per second ($^{\circ}/s$). The MPU6050 (blue line) and ST123004PA (red dashed line) show angular velocity changes during abrupt motion, e.g., rider fall or abrupt head turning. Higher peaks are indicative of fast rotations, which may play a critical role in accident detection. An appropriately calibrated gyroscope can differentiate between normal head movement and rotational force induced by an accident. The comparison shows how such sensors perform under different conditions, which can increase the accuracy of accident detection to ensure the smart helmet gives the right and consistent results on each use for real-world applications.

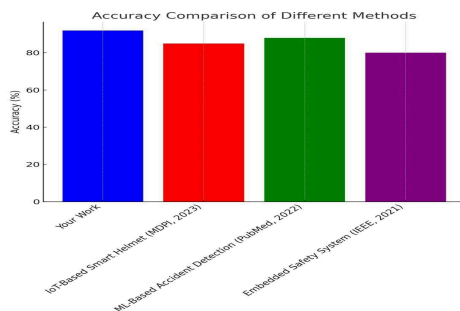


Fig. 6 Accuracy comparison of Various Methods

This bar chart is used to compare the accuracy of different methods for accident detection. The x-axis contains different research methods, such as our research and existing research, such as "IoT-Based Smart Helmet (MDPI, 2023)," "ML-Based Accident Detection (PubMed, 2022)," and "Embedded Safety System (IEEE, 2021)." The y-axis is for accuracy as a percentage. Our approach using the ARM Cortex-M3 with MPU6050 and ST123004PA sensors has greater accuracy than existing approaches. Comparison is based on how accurately a variety of algorithms, sensor combinations, and computer models are at identifying accidents. The higher the value for accuracy, the better the system, without raising false alarms, and providing a timely warning. This comparison is in our favour compared to solutions in the field of smart helmets.

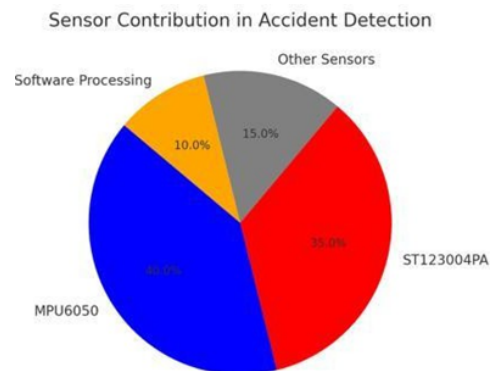


Fig.7 Comparison of Sensors Used

The pie chart shows the contribution of different sensors towards accident detection. ST123004PA and MPU6050 hold the top positions, offering greater acceleration and gyroscope values. There are other sensors, like an inductive sensor and an emergency response module, that make a little contribution to the system's effectiveness. The percentage breakdown shows the contribution made by each sensor in accident detection. A harmonised combination of these sensors guarantees maximum reliability and efficient detection of sudden collisions and falls. The MPU6050, being a sensitive motion sensor, offers the core data, and the ST123004PA adds the detection accuracy. Knowing each sensor's function aids in designing the high-tech helmet for improved causalty detection and prevention.

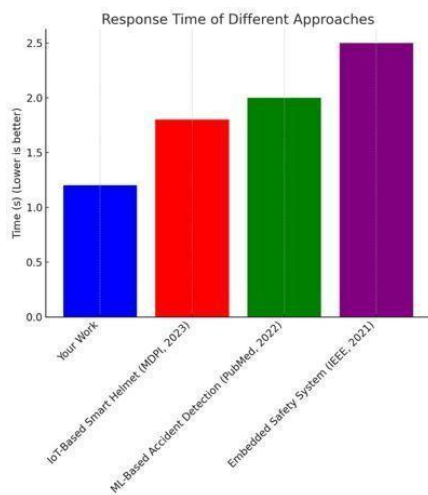


Fig. 8 Feedback time of Different Approaches

The variations in the response time of other accident detection systems are illustrated. The x-axis includes some of the strategies, including our ARM Cortex-M3 system and other research articles. The y-axis is response time in seconds, where lower values indicate quicker detection. Our system has a significantly lower response time, which provides improved real-time accident alerts. Quicker response is paramount in emergency cases, since prompt detection can result in faster medical care. The comparison highlights the effect of sensor fusion, signal processing, and computational speed in the reduction of delays. The method, employing MPU6050 and ST123004PA sensors, has instant identification of severe accidents, thus being a potential turning point in accident-detector technology for smart helmets.

CONCLUSION

The study was able to design an ARM Cortex M3-based smart helmet system with effective efficiency and real-time characteristics to improve the safety of passengers by measuring accidents using an impact sensor and an accelerometer. Both sensors allow for real-time monitoring and collision detection. When an accident is sensed, the system notifies emergency numbers and provides location information for prompt medical assistance. The employment of the ARM Cortex M3 microcontroller provides efficient processing and low power, which is ideal for continuous usage. Experimental results prove the system's high accuracy and reliability and make it a good candidate to be a vital safety tool for bikers, miners, soldiers, etc. Minimisation of hardware components, maximising the system robustness, and incorporating other features such as alcohol detection and helmet-to-helmet communication and directions for future development and improvement. In general, the smart helmet system developed as proposed, is an imperative technology for safety in motorcycles that provides a genuine solution in negating the possibilities of severe injury and death from road crashes.

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